

Constrained sphere packing and the interpolation problem for positive definite functions

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Joint work with Sujit Sakharam Damase, IISc, Bangalore

A photograph of two young men standing in front of a large chalkboard in a lecture hall. The man on the left is wearing a green t-shirt with a graphic and a black jacket, while the man on the right is wearing a light-colored patterned sweater. The chalkboard is filled with handwritten mathematical equations and diagrams. The left side of the board contains various mathematical expressions, including $X = \{x_i\}^n$, $G = \mathbb{S}_n \times (\mathbb{Z}/2\mathbb{Z})$, and $I(X) = \{1 - x_i\}$. The right side features a diagram of a graph G with nodes v_1, v_2, v_3, v_4 and edges, along with equations like $y = \bigoplus_{i=1}^n x_i$ and $\frac{1}{2} \sum_{i=1}^n x_i^2 = 10/12$. The background shows a window with a view of a building and a red carpeted floor.

2 / 29

Acknowledgments

We would like to thank:

- The Indian Institute of Science, Bangalore
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Permanently multiplying the network in all areas with an understanding of the transient nature of division is key to a continuing human role in mathematical progress. “Proof and conjecture, and keep the SF’s score low.” Paul Erdős

Sphere packing

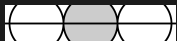
Sphere packing

Want to find the the optimal packing of spheres in some dimension.

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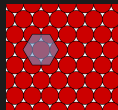
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$$n = 1$$



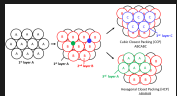
$$n = 2$$

Lagrange



$$n = 3$$

Hales (90s)



$$n = 8, 24$$

Viazovska et al. (2010s)

$$E_8,$$

Leech lattice

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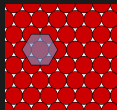
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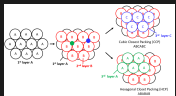
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For no other cases is the optimal density known exactly, although the optimal packings do not come from lattices.

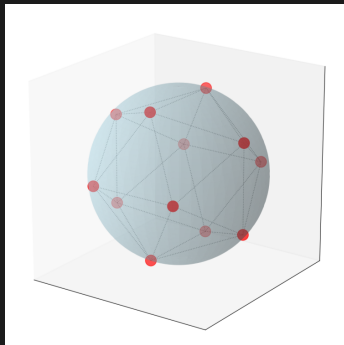
Spherical codes

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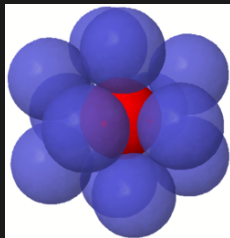
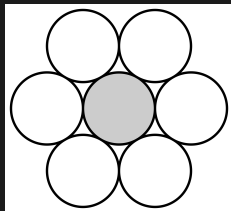
Kissing number problem

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The kissing number is equal to the maximal size of a spherical code with minimum angle $\pi/3$.

Delsarte's method

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Observation: Let $\pi/2 < \theta < \pi$. Any spherical code with minimum angle θ has cardinality less than

$$\frac{\cos \theta - 1}{\cos \theta}.$$

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Strategy: Given $n \in \mathbb{N}$ and $0 < \theta < \pi/2$ find the maximum $\tilde{\theta} > \pi/2$ such that there is a map from the sphere in n -dimensional space to the sphere in a larger dimensional space such that every spherical code with minimum angle θ is mapped to a code with minimum angle at least $\tilde{\theta}$.

Positive definite functions

Positive definite functions

Let G be group, let H be subgroup of G . A (completely) positive definite function

$$f : G/H \rightarrow \mathbb{R} \text{ (resp. self-adjoint operators)}$$

satisfies: $[f(g_i^* g_j)]_{i,j}$ is positive semidefinite for all $g_1, \dots, g_d \in G$.

Stinespring factorization

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$$f(g) = v^* \pi(g) v$$

where π is a representation of G and v is a vector which is invariant for $\pi|_H$. If f is continuous and G is a compact group, the Peter-Weyl decomposition gives this is a direct sum of irreducibles. (convex combination)

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The homomorphic analogue of **Tomiyama's theorem** gives that f is in fact bi-invariant,

$$f(h_1 g h_2) = f(g), \forall h_1, h_2 \in H, g \in G.$$

Schoenberg's theorem on positive definite functions

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If f is positive definite and continuous, $f(g) = \sum_{i=1}^{\infty} a_i C_i^{(n)}(\gamma_{11})$ where $a_i \geq 0$ and the $C_i^{(n)}$ are Gegenbauer polynomials, a natural basis for bi-invariant functions. (We will normalize to have $C_i^{(n)}(1) = 1$)
(Schoenberg '42, Bochner '41)

Operations on positive definite functions

Operations on positive definite functions

Given $C_i^{(n)}(\gamma_{11}) = v_i^* \pi_i(\gamma) v_i$, we see that

$$C_i^{(n)} C_j^{(n)} = \sum_{k=0}^{i+j} a_k C_k^{(n)}$$

where $\sum a_k = 1, a_k \geq 0$ as

$$C_i^{(n)}(\gamma_{11}) C_j^{(n)}(\gamma_{11}) = (v_i \otimes v_j)^* (\pi_i \otimes \pi_j)(\gamma) (v_i \otimes v_j)^*$$

and $\pi_i \otimes \pi_j$ may again be broken into irreducibles.

Kernel embedding

Kernel embedding

Theorem (Aronszajn's theorem)

Let X be a set. Let $A : X \times X \rightarrow \mathbb{C}$ such that $[A(x_i, x_j)]_{i,j}$ is positive semidefinite for arbitrary finite collections $x_1, \dots, x_d$.

There exists a Hilbert space $\mathcal{H}$ and $\iota_x : X \rightarrow \mathcal{H}$ such that $\langle \iota_x, \iota_y \rangle = A(x, y)$.

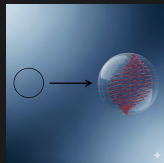
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Thus a positive definite function $f : [-1, 1] \rightarrow \mathbb{R}$ such that $f(1) = 1$ induces an embedding via the kernel $f(\langle x, y \rangle)$ which lands in the sphere of the Hilbert space as $f(\langle x, x \rangle) = 1$.



Delsarte's method

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Theorem (Delsarte '72)

Let $0 < \theta < \pi$. Let $g : [-1, 1] \rightarrow \mathbb{R}$ be a positive definite function on the sphere in n dimensions such that $g(1) = 1$ and $g(x) \leq 0$ if $x \leq \cos \theta$.

Let $\bar{g}$ the constant coefficient of g in its Gegenbauer expansion. (That is the average value of $g(\langle x, y \rangle)$ over x, y in the unit sphere. Let $\mathcal{C}$ be spherical code with minimal angle θ . Then,

$$|\mathcal{C}| \leq \frac{1}{\bar{g}}.$$

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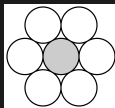
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Linear programming methods: Delsarte-Goethals-Seidel, '77 Kabitsansky-Levenshtein '78.

Kissing number in $\mathbb{R}^2$

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The polynomial

$$p(t) = \frac{4}{9}(t - 1/2)(t + 1/2)^2(1 + t)$$

satisfies

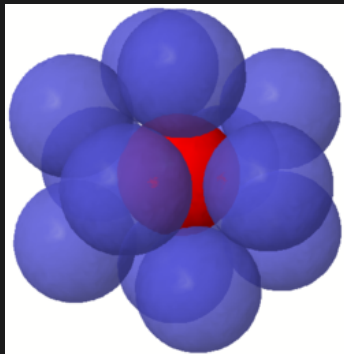
$$p(x) = \frac{1}{6}C_0^{(2)}(x) + \frac{1}{3}C_1^{(2)}(x) + \frac{5}{18}C_2^{(2)}(x) + \frac{1}{6}C_3^{(2)}(x) + \frac{1}{18}C_4^{(2)}(x)$$

$$\text{i.e. } p(\cos \theta) = \frac{1}{6} + \frac{1}{3} \cos \theta + \frac{5}{18} \cos 2\theta + \frac{1}{6} \cos 3\theta + \frac{1}{18} \cos 4\theta$$

Open Problem: What do these coefficients mean?

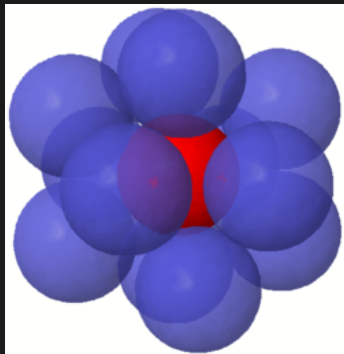
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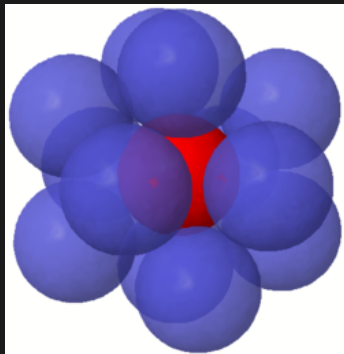
The kissing number is 12, but Delsarte gives an estimate greater than 13.

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Kissing number in $\mathbb{R}^3$



The kissing number is 12, but Delsarte gives an estimate greater than 13. Packing is loose. Musin resolves using two kernel estimates. ('06)

Delsarte sharpness

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Sharp

$n = 2$ kissing number

$n = 8, 24$ kissing number

Sloane, Odlyzko, 79

Not Sharp

$n = 3, 4$ kissing number

Musin

$X = \{-1, 1/2, -1/2, 1\}, n = 10$

Damase, P. ($\frac{1}{\bar{g}} = 46.$)

Cohn-Elkies method

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Theorem (Cohn-Elkies, '03)

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is an admissible function, is not identically zero, and satisfies the following two conditions:

- 1 $f(x) \leq 0$ for $|x| \geq 1$, and
- 2 $\hat{f}(t) \geq 0$ for all t .

Then the center density of n -dimensional sphere packings is bounded above by

$$\frac{f(0)}{2^n \hat{f}(0)}.$$

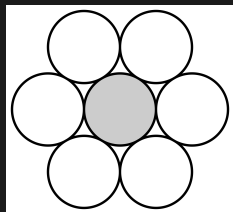
Constrained angle codes

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For example, the spherical code given by six equispaced points on the circle has cosines in $\{0, -1, 1/2, -1/2\}$.

Interpolation for positive definite functions

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c.f. Krein, Jorgensen, etc. on similar but different interpolation problems. ($\Gamma - \Gamma$, not necessarily symmetric, only ask for f to induce a kernel on Γ .)

Arveson extension and pathological Class I representations

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Theorem (Damase-P.)

$SO(\infty)/1 \oplus SO(\infty)$ and $SO(2)/\{1\}$ are not pathological.

Open Problem: $SO(n)/SO(n-1)$? Open Problem: Pathological Delsarte estimates arising from nonmeasurable pd functions.

Continuous extension problem

Continuous extension problem

Let X be a finite subset of $[-1, 1]$. Let $f : X \rightarrow \mathbb{R}$. When is there a continuous positive definite function on the sphere in n -dimensions $\tilde{f} : [-1, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}|_X = f$?

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Theorem (Damase, P.)

Let f correspond to a solvable continuous extension problem. There is a polynomial solution $\tilde{f}$.

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Viazovska et al. functions for Cohn-Elkies are modular functions. (Low algebraic complexity. Open problem: Uniqueness.)

Delsarte hallucinations

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- Consider $X = \{-1, 1/2, -1/2, 1\}$, $n = 10$. There is no corresponding constrained angle code, but we have $\frac{1}{g} = 46$ is an integer. Many such examples exist of **Delsarte hallucinations**.

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- On the other hand, the example $X = \{1/3, -1/3, 1\}$ has a spherical code with 10 points in the 5 dimensional sphere, but the sign pattern comes from a Petersen graph.
- Given an X , determining if there exists a subset of the n -sphere with cosines in X of some size k is a difficult combinatorial matrix completion problem.

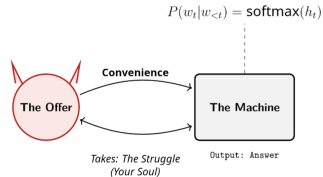
Open problem: Are Delsarte hallucinations a coincidence?

The future is now

Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvelous machine.



—Sir Michael Atiyah, 2002

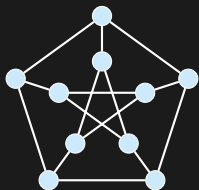


AI is the offer made by the devil to the mind.

The devil says: I will give you this powerful machine, it will complete any thought you like.

All you need to do is give me your soul: give up **the struggle** and you will have this marvelous machine.

— Gemini, 2025



The Petersen Graph

Seidel Matrix S

$$\begin{pmatrix} 0 & - & - & 1 & 1 & - & 1 & 1 & 1 & 1 \\ - & 0 & - & - & 1 & 1 & - & 1 & 1 & 1 \\ - & - & 0 & - & - & 1 & 1 & - & 1 & 1 \\ 1 & - & - & 0 & - & 1 & 1 & 1 & - & 1 \\ 1 & 1 & - & - & 0 & 1 & 1 & 1 & 1 & - \\ - & 1 & 1 & 1 & 1 & 0 & 1 & - & 1 & - \\ 1 & - & 1 & 1 & 1 & 1 & 0 & 1 & - & - \\ 1 & 1 & - & 1 & 1 & - & 1 & 0 & - & - \\ 1 & 1 & 1 & - & 1 & 1 & - & - & 0 & - \\ 1 & 1 & 1 & 1 & - & - & - & - & - & 0 \end{pmatrix}$$

Spectrum: $3^{(5)}, -3^{(5)}$

Gram Matrix G

Construct $X = \{1, \pm 1/3\}$:

$$G = I + \frac{1}{3}S \succeq 0$$

Discovery: AI = 爱

- Example suggested by **Gemini**.
- Verified & Formalized by **Damase & P.**

Soft Thresholding

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- 1 Given an empirical correlation matrix of rank N , (low sample statistics) one may want to threshold small entries to 0 to obtain sparsity. However, such a process does not preserve the positive definiteness, as it is not a positive definite function.

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Theorem (Damase, P.)

The faithfulness constant of $[-\varepsilon, \varepsilon]$ does not go to 1 if $N \geq 4$.

Open problem: What do the latent spaces mean?

Soft Thresholding

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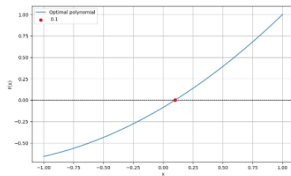


FIGURE 1. The thresholding function for $K = \{0.1\}$, $n = 3$

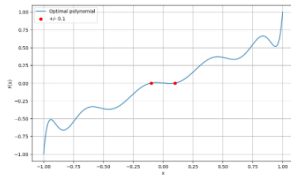


FIGURE 2. The thresholding function for $K = \{\pm 0.1\}$, $n = 3$

